Welcome to Isaac Lab!

Contents

Isaac Lab is a unified and modular framework for robot learning that aims to simplify common workflows in robotics research (such as reinforcement learning, learning from demonstrations, and motion planning). It is built on NVIDIA Isaac Sim to leverage the latest simulation capabilities for photo-realistic scenes, and fast and efficient simulation.

The core objectives of the framework are:

- **Modularity**: Easily customize and add new environments, robots, and sensors.
- **Agility**: Adapt to the changing needs of the community.

- **Openness**: Remain open-sourced to allow the community to contribute and extend the framework.
- **Batteries-included**: Include a number of environments, sensors, and tasks that are ready to use.

Key features available in Isaac Lab include fast and accurate physics simulation provided by PhysX, tiled rendering APIs for vectorized rendering, domain randomization for improving robustness and adaptability, and support for running in the cloud.

Additionally, Isaac Lab provides a variety of environments, and we are actively working on adding more environments to the list. These include classic control tasks, fixed-arm and dexterous manipulation tasks, legged locomotion tasks, and navigation tasks. A complete list is available in the environments section.

Isaac lab is developed with specific robot assets that are now **Batteries-included** as part of the platform and are ready to learn! These robots include...

- **Classic** Cartpole, Humanoid, Ant
- **Fixed-Arm and Hands**: UR10, Franka, Allegro, Shadow Hand
- **Quadrupeds**: Anybotics Anymal-B, Anymal-C, Anymal-D, Unitree A1, Unitree Go1, Unitree Go2, Boston Dynamics Spot
- **Humanoids**: Unitree H1, Unitree G1
- **Quadcopter**: Crazyflie

The platform is also designed so that you can add your own robots! Please refer to the How-to Guides section for details.

For more information about the framework, please refer to the technical report [MRT+25]. For clarifications on NVIDIA Isaac ecosystem, please check out the Isaac Lab Ecosystem section.

The Isaac Lab framework is open-sourced under the BSD-3-Clause license, with certain parts under Apache-2.0 license. Please refer to License for more details.

If you use Isaac Lab in your research, please cite our technical report:

```
@article{mittal2025isaaclab,
   title={Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Rc
   author={Mayank Mittal and Pascal Roth and James Tigue and Antoine Richard a
   journal={arXiv preprint arXiv:2511.04831},
   year={2025},
   url={https://arxiv.org/abs/2511.04831}
}
```

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Isaac Lab

Getting Started

Overview

Features